

Research Proposal

Reproducing and Adapting Enhanced Super-Resolution Generative Adversarial Networks (ESRGAN)

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1 Introduction

Generative Adversarial Networks (GANs) have significantly advanced the field of Single Image Super-Resolution (SISR). Building upon the foundation of SRGAN, the Enhanced SRGAN (ESRGAN) framework, introduced by Wang et al. (2018), pushes the boundaries of visual quality. By introducing Residual-in-Residual Dense Blocks (RRDB) and a relativistic discriminator, ESRGAN generates highly realistic textures while mitigating the unpleasant artifacts often produced by standard GANs.

This project aims to reproduce the results of the original ESRGAN paper using an existing robust PyTorch implementation, and introduce a controlled modification by shifting the training data distribution to evaluate the model's adaptability, robustness, and potential for domain-specific hallucination errors.

2 Background and Objectives

Objective:

- Reproduce the ESRGAN training pipeline and baseline results using a reliable open-source implementation.
- Modify the experimental setup by replacing the standard, highly diverse training dataset with a strictly domain-specific dataset (e.g., satellite imagery or medical scans).

- Compare the performance, visual artifacts, and metric scores against the baseline models to understand the impact of the RRDB architecture on specialized domains.

3 Proposed Method and Methodology

Dataset

Instead of using the standard benchmark datasets like DIV2K and Flickr2K (which contain diverse, general-purpose high-resolution images), we will replace the training data with a domain-specific dataset such as high-resolution Satellite Imagery (e.g., DOTA or SpaceNet patches). This will allow us to study the impact of dataset variance on the generator’s ability to hallucinate details where mathematical accuracy is critical.

Baseline

The original ESRGAN implementation, featuring the RRDB-based generator and a relativistic discriminator, trained on standard broad datasets. We will rely on a validated open-source PyTorch pipeline (e.g., Lornatang’s implementation) to establish this baseline.

Proposed Approach

We will fine-tune or train the ESRGAN architecture entirely from scratch on our chosen specialized dataset. By doing so, we aim to analyze the representation quality and identify how the model adapts its texture generation when constrained to a single specific visual domain (e.g., avoiding the hallucination of urban textures in natural landscapes).

Evaluation Protocol

- **Quantitative analysis:** Using standard SISR metrics on an unseen test set from the same domain.
- **Qualitative analysis:** Visual inspection of generated artifacts, structural integrity, and high-frequency details.

Validation and Comparison

We will compare the outputs of our specialized model against those of a generic pre-trained ESRGAN model. We will evaluate whether a domain-specific dataset improves the realism for that specific category of images, or if it introduces unexpected artifacts (e.g., fabricating structures that do not exist).

Metrics

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index Measure)
- Perceptual Index (PI) / LPIPS
- Training time and computational cost comparison

4 Expected Results

We expect that training ESRGAN on a specialized dataset will yield superior qualitative results for images within that specific domain compared to the generalist baseline. However, because ESRGAN’s architecture is highly prone to “hallucinating” realistic but fake details, we anticipate highlighting significant limitations when applying the model to fields requiring strict structural fidelity, such as satellite mapping.

5 Conclusion

This work will validate the ESRGAN architecture while extending its application to specialized data regimes. By comparing the results with the original study, we will provide concrete insights into how data distribution directly affects the perceptual quality and the reliability of relativistic Generative Adversarial Networks in complex super-resolution tasks.

6 References

- Wang, X., Yu, K., Wu, S., Gu, J., Liu, Y., Dong, C., Loy, C. C., Qiao, Y., & Tang, X. (2018). *ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks*. arXiv:1809.00219. <https://arxiv.org/abs/1809.00219>